From target states to two common reports

One joint law; w=(3/5,2/5), a=(3/4,1/4), τ=1/5; M=6.

A1. State 1: probability 3/4

Unit	θ_1	θ_2	Composite c
A	4	4	4
B	3	2	11/4
C	2	3	9/4
D	1	1	1

A2. State 2: probability 1/4

Unit	θ_1	θ_2	Composite c
A	3	1	5/2
B	4	2	7/2
C	2	3	9/4
D	1	4	7/4

$c_i = (3/4)\theta_{i1} + (1/4)\theta_{i2}$

w weights errors; a combines target magnitudes.

B. Forward error coefficients

	AB	AC	AD	BC	BD	CD
Weighted direction	1/4	1/10	1/10	2/5	1/10	1/10
Composite	1/4	0	0	0	0	0

C. Weighted-direction optimum 1223

A	B C	D

g = (1,2,2,3); N = 5

AB	AC	AD	BC	BD	CD
●	●	●	○	●	●

C. Composite optimum 1123

A B	C	D

g = (1,1,2,3); N = 5

AB	AC	AD	BC	BD	CD
○	●	●	●	●	●

D. Same N = 5; evaluate each loss

Report	Weighted risk	Composite risk
1223	**17/120**	3/40
1123	1/6	**1/30**

Filled: forward direction. Open: withhold. Both disclose five of six pairs.
Risk uses M=6; compare minima within each target, not between target definitions.